Kyle Chiem

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Education

The Pennsylvania State University – Schreyer Honors College University Park, PA
Bachelor of Science in Computer Science and Math, College of Engineering *Aug. 2024 - May 2028*

Cumulative GPA: 4.00/4.00

Relevant Coursework: Data Structures & Algorithms, Systems Programming, Object-Oriented Programming, Digital Design

Experience

Adobe May 2026 - Aug. 2026
Incoming Software Development Engineer *San Jose, CA*

- Selected for the Engagement Services and Products team to develop full-stack **React.js** and **Node.js** platforms that facilitates **Agent-to-Agent** (A2A) workflows and integrate **Model Context Protocol** (MCP) on **AWS** infrastructure.

HackPSU Tech Team Aug. 2025 - Present
Software Engineer *University Park, PA*

- Contribute to full-stack **Next.js (React, TypeScript)** platform supporting **1000+ participants** during hackathon
- Standardize API development by implementing **Role-Based Access Control (RBAC)** and **Swagger (OpenAPI) documentation**, using DTO-based validation to ensure **99% schema accuracy** across all hackathon services
- Engineered **NestJS** room reservation API, implementing **validation logic** and **asynchronous conflict detection** providing continuous data integrity, reducing workspace discovery time upwards of **10 minutes** to **less than 10 seconds**

The Pennsylvania State University Feb 2025 - Sep. 2025
Research Intern - Robotics & Deep Learning *University Park, PA*

- Developed **reinforcement learning** locomotion controllers on **Isaac Lab** to observe humanoid robot stability metrics
- Leveraged **imitation learning** using proprietary **motion capture data** to pretrain, reducing simulation steps by 50%
- Executed **4000+ parallel simulations**, automating pipelines with **PyTorch** and evaluation with **TensorBoard**
- Integrated **domain randomization** and **sensor noise modeling**, making controller robust for in-person deployment

Nittany AI Alliance Student Chapter Jan. 2025 - May 2025
Machine Learning Apprentice *University Park, PA*

- Designed a **RAG Chatbot** using Sentence Transformers and **Tiny LLaMa LLM** for finance-related Q&A over PDFs
- Preprocessed **100,000+ NIH Chest X-ray** data to train a **CNN** to detect pneumonia patients at an accuracy of **92%**
- Explored overfitting control, RL reward shaping, and model evaluation with precision and confusion matrix analysis

Projects

Surviv (HooHacks Spring '26 Winner) | *Swift, Mesh Networking, PyTorch* | *GitHub* Mar. 2026 - Present

- Engineered **decentralized P2P mesh network**, featuring message de-duplication and enabling off-grid safety alerts
- Trained **lightweight CNN** for low-power on-device threat detection, achieving **80.9% accuracy and 94.6% recall**
- Incorporated **A* pathfinding** on dynamic graphs from street data, navigating users to safe zones while avoiding threats

Rubik's Cube Simulator | *TypeScript, Vue.js, Three.js, GSAP* | *GitHub* Oct. 2025 - Present

- Architectured a 3D puzzle engine with **Vue.js** and **Three.js**, mapping reactive vectors directly to dynamic meshes
- Implemented **mutex-based locking** within the **GSAP** pipeline to prevent race conditions from concurrent user inputs
- Built a Rubik's Cube group-theory state manager, tracking over **43 quintillion permutations** with O(1) validation

Tetris AI | *Python, PyTorch, TensorBoard, NumPy* | *GitHub* May 2025 - Oct. 2025

- Designed a **deep RL network** with **Python and PyTorch** to learn Tetris, clearing **800+ lines** per game
- Vectorized matrix operations using **NumPy**, achieving over **3x speedup** and reducing **100+ lines of code**
- Implemented **prioritized experience replay** using **segment trees** to improve sample efficiency and stabilize training
- Leveraged **TensorBoard** in analyzing performance metrics to improve overall model structure and hyperparameters

Technical Skills

Languages: Python, TypeScript, Java, C++, C, SQL, JavaScript, Kotlin, Flutter
Technologies: FastAPI, SQLAlchemy, NestJS, Node.js, AWS, PostgreSQL, Git, PyTorch, Firebase
Concepts: Backend System Design, Reinforcement Learning, Scalable Infrastructure, API Integration, Full-Stack Development, Problem Solving, Analytical Reasoning